

Operating Systems

CSIT — 4th Semester — 2080 — Answer Sheet
Subject Code: CSC264

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. How DMA operation is performed? Consider a disk with 200 tracks and the queue has random requests from different processes in the order: 45, 48, 29, 17, 80, 150, 28, and 188. Find the seek time using FIFO, SSTF and SCAN. Assume that the initial position of head is at 100.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How do you distinguish between deadlock and starvation? Describe the necessary conditions for deadlock. Explain the working mechanism of TLB.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Why do we need to schedule process? Find the average waiting time and average turnaround time for the following set of processes using FCFS, SJF, RR(Quantum: 3) and Shortest Remaining Time Next.
Process CPU Burst Time Arrival Time
P1 200 P2 25 P3 15 P4 30 P5 45

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. What is system call? Describe the transition between different states of a process.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Discuss about contiguous and linked list allocation approach in implementing files.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Why do we need virtual memory? Describe the structure of a page table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Illustrate the term safe and unsafe state in deadlock prevention with a scenario.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How lock variable is used in achieving mutual exclusion? Describe.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Why do we need hierarchical directory system? Explain the structure of a disk.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Find the number of page faults using FIFO and LRU for the reference string, 4, 3, 6, 1, 7, 6, 1, 2, 1, 2 with frame size 3.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define working set. How does clock page replacement algorithm work?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on a. Inodes b. RAID

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.